Python Programming Introduction

Learning notes for document-loading and RAG practice

What is Python?

Python is a high-level, interpreted programming language. It is known for readable syntax and a large ecosystem of libraries. Python was created by Guido van Rossum and first released in 1991.

Key Features

Python is easy to learn, supports multiple programming styles, runs on different operating systems, and has a large standard library. It also has a strong developer community.

Common Data Types

Common built-in data types include integers, floating-point numbers, strings, booleans, lists, tuples, sets, and dictionaries. A list stores an ordered collection that can be changed. A tuple is ordered but immutable. A dictionary stores key-value pairs.

Functions and Modules

A function is a reusable block of code that performs a task. Functions help organize programs and reduce repetition. A module is a Python file containing reusable code, such as functions and classes.

Applications

Python is widely used in web development, automation, data analysis, scientific computing, machine learning, and artificial intelligence.